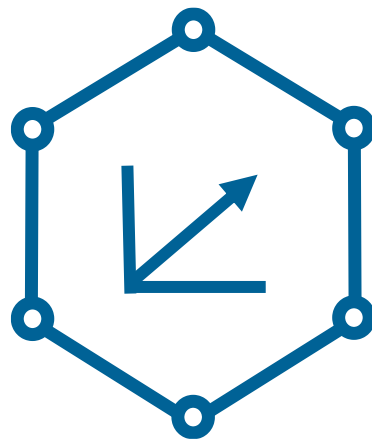
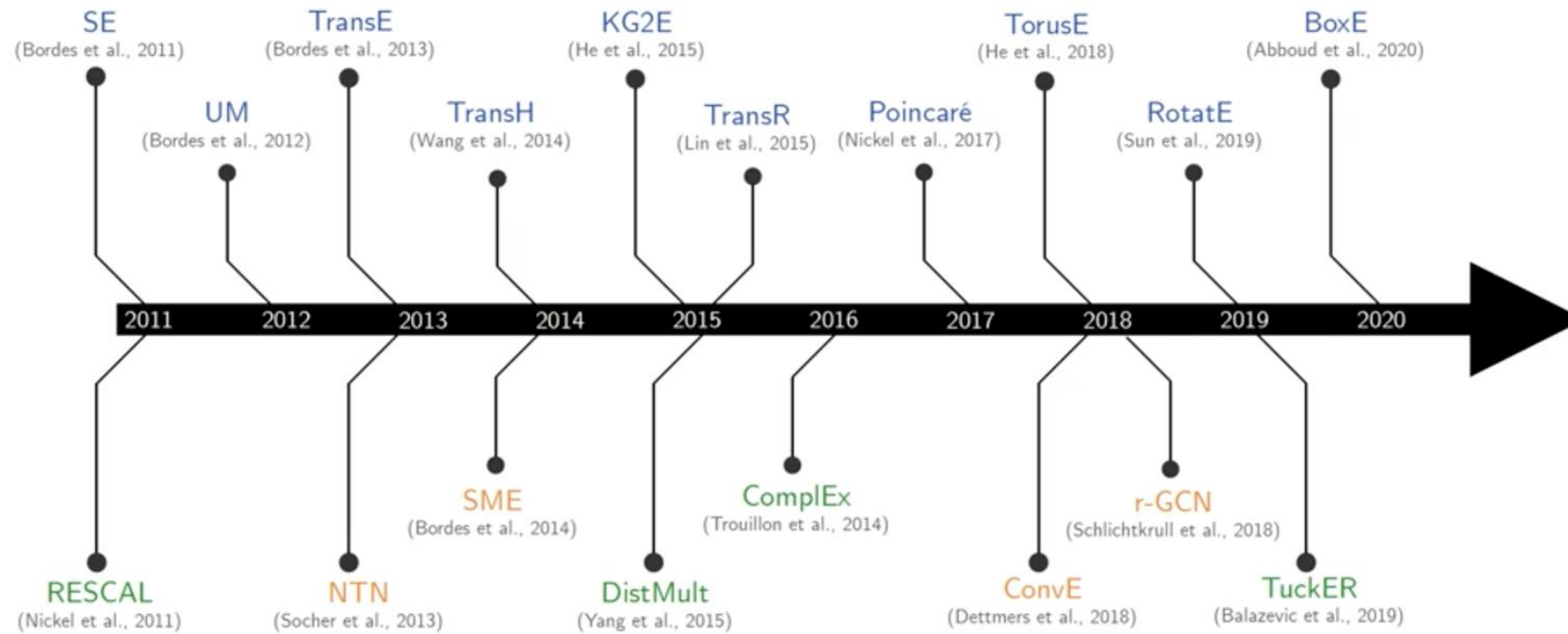




Knowledge Graph Embeddings A Glimpse Beyond

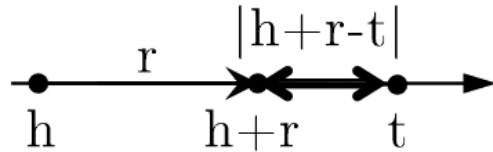
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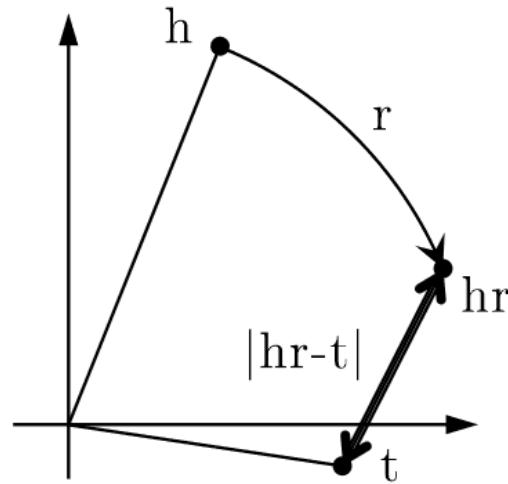


Courtesy of Ismail Ilkan Ceylan

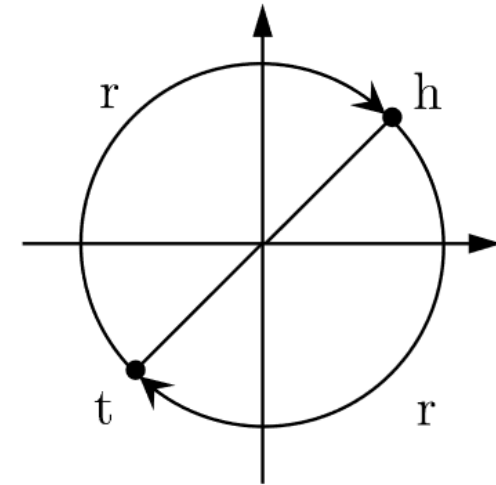
RotatE



(a) TransE models r as translation in real line.



(b) RotatE models r as rotation in complex plane.



(c) RotatE: an example of modeling symmetric relations r with $r_i = -1$

Score Functions

Model	Score Function	
SE (Bordes et al., 2011)	$-\ \mathbf{W}_{r,1}\mathbf{h} - \mathbf{W}_{r,2}\mathbf{t}\ $	$\mathbf{h}, \mathbf{t} \in \mathbb{R}^k, \mathbf{W}_{r,\cdot} \in \mathbb{R}^{k \times k}$
TransE (Bordes et al., 2013)	$-\ \mathbf{h} + \mathbf{r} - \mathbf{t}\ $	$\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$
TransX	$-\ g_{r,1}(\mathbf{h}) + \mathbf{r} - g_{r,2}(\mathbf{t})\ $	$\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$
DistMult (Yang et al., 2014)	$\langle \mathbf{r}, \mathbf{h}, \mathbf{t} \rangle$	$\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$
ComplEx (Trouillon et al., 2016)	$\text{Re}(\langle \mathbf{r}, \mathbf{h}, \bar{\mathbf{t}} \rangle)$	$\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{C}^k$
HolE (Nickel et al., 2016)	$\langle \mathbf{r}, \mathbf{h} \otimes \mathbf{t} \rangle$	$\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$
ConvE (Dettmers et al., 2017)	$\langle \sigma(\text{vec}(\sigma([\bar{\mathbf{r}}, \bar{\mathbf{h}}] * \mathbf{\Omega}))\mathbf{W}), \mathbf{t} \rangle$	$\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$
RotatE	$-\ \mathbf{h} \circ \mathbf{r} - \mathbf{t}\ ^2$	$\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{C}^k, r_i = 1$

Expressive Power

Model	Score Function	Symmetry	Antisymmetry	Inversion	Composition
SE	$-\ \mathbf{W}_{r,1}\mathbf{h} - \mathbf{W}_{r,2}\mathbf{t}\ $	$\times$	$\times$	$\times$	$\times$
TransE	$-\ \mathbf{h} + \mathbf{r} - \mathbf{t}\ $	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
TransX	$-\ g_{r,1}(\mathbf{h}) + \mathbf{r} - g_{r,2}(\mathbf{t})\ $	$\checkmark$	$\checkmark$	$\times$	$\times$
DistMult	$\langle \mathbf{h}, \mathbf{r}, \mathbf{t} \rangle$	$\checkmark$	$\times$	$\times$	$\times$
Complex	$\text{Re}(\langle \mathbf{h}, \mathbf{r}, \bar{\mathbf{t}} \rangle)$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$
RotatE	$-\ \mathbf{h} \circ \mathbf{r} - \mathbf{t}\ $	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

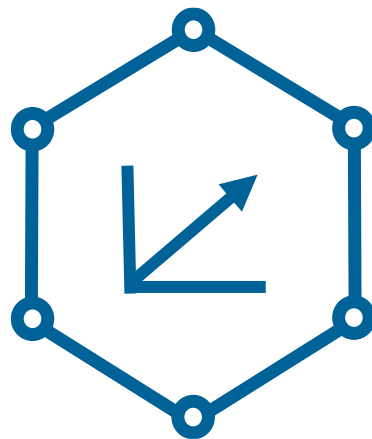
Datasets (25)

The citation for each dataset corresponds to either the paper describing the dataset, the first paper published using the dataset with knowledge graph embedding models, or the URL for the dataset if neither of the first two are available.

Name	Documentation	Citation	Entities	Relations
Clinical Knowledge Graph	pykeen.datasets.CKG	Santos et al., 2020	7617419	11
CoDEx (large)	pykeen.datasets.CoDExLarge	Safavi et al., 2020	77951	69
CoDEx (medium)	pykeen.datasets.CoDExMedium	Safavi et al., 2020	17050	51
CoDEx (small)	pykeen.datasets.CoDExSmall	Safavi et al., 2020	2034	42
ConceptNet	pykeen.datasets.ConceptNet	Speer et al., 2017	28370083	50
Countries	pykeen.datasets.Countries	ZhenfengLei/KGDatasets	271	2
Commonsense Knowledge Graph	pykeen.datasets.CSKG	Ilievski et al., 2020	2087833	58
DB100K	pykeen.datasets.DB100K	Ding et al., 2018	99604	470
DBpedia50	pykeen.datasets.DBpedia50	Shi et al., 2017	24624	351
Drug Repositioning Knowledge Graph	pykeen.datasets.DRKG	gnn4dr/DRKG	97238	107
FB15k	pykeen.datasets.FB15k	Bordes et al., 2013	14951	1345
FB15k-237	pykeen.datasets.FB15k237	Toutanova et al., 2015	14505	237
Hetionet	pykeen.datasets.Hetionet	Himmelstein et al., 2017	45158	24
Kinships	pykeen.datasets.Kinships	ZhenfengLei/KGDatasets	104	25
Nations	pykeen.datasets.Nations	ZhenfengLei/KGDatasets	14	55
OGB BioKG	pykeen.datasets.OGBBioKG	Hu et al., 2020	45085	51
OGB WikiKG	pykeen.datasets.OGBWikiKG	Hu et al., 2020	2500604	535
OpenBioLink	pykeen.datasets.OpenBioLink	Breit et al., 2020	180992	28
OpenBioLink (F1)	pykeen.datasets.OpenBioLinkF1	PyKEEN/pykeen-openbiolink-benchmark	116425	19
OpenBioLink (F2)	pykeen.datasets.OpenBioLinkF2	PyKEEN/pykeen-openbiolink-benchmark	110628	17

Models (25)

Name	Reference	Citation
ComplEx	pykeen.models.ComplEx	Trouillon et al., 2016
ComplExLiteral	pykeen.models.ComplExLiteral	Kristiadi et al., 2018
ConvE	pykeen.models.ConvE	Dettmers et al., 2018
ConvKB	pykeen.models.ConvKB	Nguyen et al., 2018
DistMult	pykeen.models.DistMult	Yang et al., 2014
DistMultLiteral	pykeen.models.DistMultLiteral	Kristiadi et al., 2018
ERMLP	pykeen.models.ERMLP	Dong et al., 2014
ERMLPE	pykeen.models.ERMLPE	Sharifzadeh et al., 2019
HolE	pykeen.models.HolE	Nickel et al., 2016
KG2E	pykeen.models.KG2E	He et al., 2015
MuRE	pykeen.models.MuRE	Balažević et al., 2019
NTN	pykeen.models.NTN	Socher et al., 2013
PairRE	pykeen.models.PairRE	Chao et al., 2020
ProjE	pykeen.models.ProjE	Shi et al., 2017
RESCAL	pykeen.models.RESCAL	Nickel et al., 2011
RGCN	pykeen.models.RGCN	Schlichtkrull et al., 2018
RotatE	pykeen.models.RotatE	Sun et al., 2019
Simple	pykeen.models.Simple	Kazemi et al., 2018
StructuredEmbedding	pykeen.models.StructuredEmbedding	Bordes et al., 2011
TransD	pykeen.models.TransD	Ji et al., 2015
TransE	pykeen.models.TransE	Bordes et al., 2013



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